# Gaspard Nordin

French engineering student at Arts et Métiers (Diplôme d'Ingénieur + MSc KIMP in Industrial Engineering), specialising in the modelling and simulation of production systems. Seeking a 6-month R&D internship in industrial engineering from February 2027 (master thesis).

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## EDUCATION

**Arts et Métiers Institute of Technology (ENSAM)** — Metz & Paris, France **Sep 2024 – Sep 2027**
*Diplôme d'Ingénieur (Master's-level engineering degree) + MSc KIMP in Industrial Engineering (research master), double degree*
Arts et Métiers, a founding school of ParisTech, is renowned for its excellence in mechanical and industrial engineering.

- MSc KIMP, Design & Manufacturing track (Paris, Sep 2026 – Sep 2027): integrated design & manufacturing, CAD/CAM, AI for design, mechatronics, manufacturing process management, robust design & big data, decision & risk analysis.
- Engineering core curriculum: Data Science, Mechanics, Thermodynamics, Materials Science, Supply Chain.
- Exchange semester at Universiti Teknologi PETRONAS, Malaysia (Mar 2026 – Aug 2026): oilfield and process-related corrosion, condition-based monitoring, mechatronics, control systems, building air-conditioning systems.

**Lycée Lakanal** — Sceaux, France **Sep 2021 – Jul 2024**
*Intensive preparatory classes (CPGE), Mathematics & Physics (MP)*

- Admitted to Arts et Métiers through the French national competitive entrance examination.

## PROFESSIONAL EXPERIENCE

**Chantiers Modernes Construction (VINCI)** — Paris, France **Jun 2025 – Jul 2025**
Site Operations Intern (worker internship) — Formwork

- Worked on the interfaces between the EOLE extension (RER E) and Line 15 of the Grand Paris Express.
- Full prepare–formwork–pour–strip cycle for reinforced concrete: fibre-reinforced segments, walls, columns and a foundation raft in a shaft.
- Read technical drawings, took measurements with surveyor support and ran visual quality checks.
- Coordinated with waterproofing and rebar teams on sequencing and site readiness.

**Tutoring London** — Remote **Oct 2024 – Present**
Private Tutor

- Prepare high-school students for the scientific baccalauréat (mathematics and physics).

## SELECTED PROJECTS

**Research Project — Inclusive Task Allocation** **Sep 2025 – Dec 2025**
Team of 5 — built the modelling framework (phase 1)
*Objective: Compare task-allocation strategies in a workshop where operators differ in capability and fatigue, on both performance and fairness.*

- Concluded that no allocation rule is optimal on every indicator: rules that maximise throughput increase fatigue and workload imbalance, while fairness-oriented rules can lower performance.
- Built a hybrid model (M/M/1 arrival queue, supervisor agent, bounded fatigue law, stochastic Petri nets, multi-agent architecture), since an exact Markov model would exceed $10^9$ states.
- Defined the performance indicators and four allocation strategies; the team then ran numerical simulations and an AnyLogic implementation.

**Statistics Project — Uncertain Catapult: Robust Inverse Design** **Dec 2025**
Team of 5 — led data processing
*Objective: Find the settings that make a noisy catapult hit a target range of 1.95 m ± 0.10 m.*

- Reached 25/25 test shots within tolerance with a robust Nelder–Mead optimisation, against 76% for plain distance-to-target minimisation.
- Validated the measurement system with a gauge R&R study (ANOVA): repeatability and reproducibility below 0.03% of total variability.
- Modelled the range with full factorial, fractional and Taguchi designs (L25 model: $R^2 = 0.97$); compared with kriging + genetic algorithm.

**Programming Project — Automated Warehouse Simulation** **Mar 2025 – May 2025**
Team of 2 — wrote the simulation code
*Objective: Simulate an automated warehouse where robots, ejectors and a conveyor move products to the exit under a central coordinator.*

- Designed the object model (one class per entity, a Robot base class with two subclasses, a coordinator validating every action).
- Implemented nearest-target assignment (Manhattan distance) and breadth-first-search path finding with collision avoidance.
- Loaded warehouse configurations from XML; produced a movement log, a usage heatmap and travel statistics per robot.

**TIPE — Markov Chains: Queues & Monopoly** **Sep 2023 – Jun 2024**
Individual project — TIPE: supervised research project in CPGE, Lycée Lakanal
*Objective: Use Markov processes to size queueing systems, then model the game of Monopoly.*

- Showed that pooling two queues into one shared M/M/2 queue cuts the mean time in the system by 24%.
- Derived the M/M/1 steady state and performance indicators; validated them with a Python simulator (10 s in theory, 9.998 s simulated).
- Modelled Monopoly as a 120-state Markov chain to compute visit probabilities and payback times, showing three houses is the optimum.

**Innovation Project — Modular Electric Motorcycle** **Oct 2024 – Jan 2025**
Team of 3 — state-of-the-art study and final report
*Objective: Identify an innovation that would help the electric motorcycle segment take off.*

- Proposed six modular systems; the five main modules cover over 90% of the needs identified.
- Linked 10 uses, 23 usage constraints and 20 technical characteristics through QFD-style weighted matrices, calibrated on market benchmarks.

## SKILLS AND INTERESTS

- Languages: **French (Native), English (C1), Spanish (B1)**, Chinese (Beginner)
- Modelling: Markov chains, queueing theory, stochastic Petri nets, multi-agent simulation, design of experiments (Taguchi), ANOVA, robust optimisation, kriging, genetic algorithms
- Software: Python (NumPy, pandas, SciPy, Matplotlib, object-oriented programming), AnyLogic, Excel, LaTeX, Microsoft Office
- Sports and Leisure: **Tennis** (nationally ranked, 15/1 in the French ranking), **Football** (5 years of competition), **Scuba diving** (PADI certified)
- Readings: philosophy and current affairs